

Student Name: Tiffany Ho
13609403

2003
TRIAL HIGHER SCHOOL CERTIFICATE

MATHEMATICS
Extension 2



General Instructions

Reading Time: 5 minutes
Working Time: 3 hours

- Attempt all questions
- Start each question on a new page
- Each question is of equal value
- Show all necessary working.
- Marks may be deducted for careless work or incomplete solutions
- Standard integrals are printed on the last page
- Board-approved calculators may be used
- This examination paper must not be removed from the examination room

Question 1. (15 marks) Start a new page.

Marks

a) Find $\int \sec^2 x (\tan^2 x + 2) dx$. 2

b) Find $\int \frac{5}{x^2 + 6x + 13} dx$. 2

c) Use $t = \tan\left(\frac{x}{2}\right)$ to find $\int \frac{dx}{1 + \sin x + \cos x}$. 3

d) Find $\int \frac{e^{2x}}{(e^x + 1)^2} dx$ using the substitution $u = e^x + 1$ 2

e) Find $\int 3^x dx$. 1

f) i) Let $I_n = \int_0^1 x^n e^x dx$ where $n \geq 0$. Show that 3

$$I_n = e - nI_{n-1} \text{ for } n \geq 1.$$

ii) Hence evaluate $\int_0^1 x^3 e^x dx$. 2

Question 2. (15 marks) Start a new page

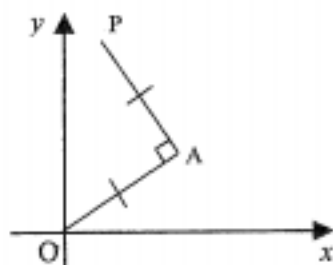
Marks

- a) Let $z = 3 - 4i$ and $w = 2 + 5i$. Express the following in the form $x + iy$, where x and y are real numbers.

- i) z^2 1
- ii) $i^3 \frac{z}{w}$ 2

- b) Find all the complex numbers $z = a + ib$, where a and b are real, such that $|z^2| + i\bar{z} = 11 + 3i$ 3

c)



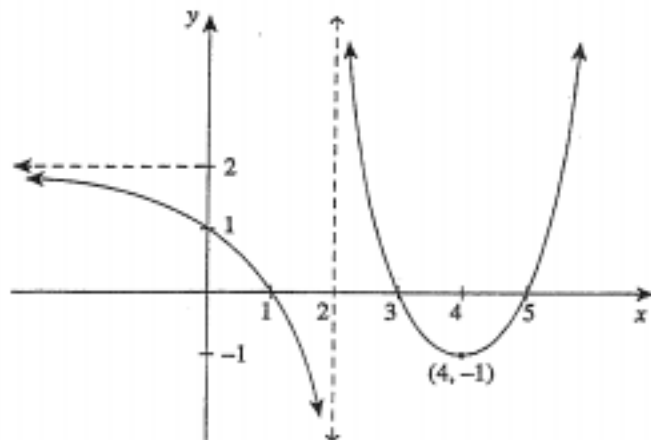
The point A in the complex plane corresponds to the complex number z . The triangle OAP is a right angled isosceles triangle.

- i) Find in terms of z the complex number corresponding to the point P . 1
- ii) Let M be the midpoint of OP . What complex number corresponds to M ? 1
- d) i) Express $3 - 3i$ in modulus-argument form. 1
- ii) Hence evaluate $(3 - 3i)^7$, expressing it in the form $a + ib$ where a and b are real numbers. 2
- e) i) On the same diagram, draw a neat sketch of the locus specified by:
 $\alpha) |z - (5 + 4i)| = 4$
 $\beta) |z + 4| = |z - 6|$ 2
- ii) Hence write down the value of z which simultaneously satisfies $|z - (5 + 4i)| = 4$ and $|z + 4| = |z - 6|$ 1
- iii) Use your diagram in (i) to determine the value(s) of k for which the simultaneous equations $|z - (5 + 4i)| = 4$ and $|z - 4i| = k$ have exactly one solution for z . 1

Question 3. (15 marks) Start a new page.

Marks

- a) The graph of $y = f(x)$ is drawn below.



As $x \rightarrow -\infty$, $f(x) \rightarrow 2$. The line $x = 2$ is a vertical asymptote. The y -intercept is $y = 1$ and the x -intercepts are $x = 1$, $x = 3$ and $x = 5$.

Draw separate *half-page* sketches of the graphs of the following:

- | | | |
|------|---|---|
| i) | $y = f(x) $ | 2 |
| ii) | $y = f(x)$ | 2 |
| iii) | $y = \frac{1}{f(x)}$ | 2 |
| iv) | $y = \tan^{-1}[f(x)]$ | 2 |
| b) | i) Find the coordinates and the nature of the stationary points on the curve $y = x^3 + 6x^2 + 9x + k$ where k is real. | 2 |
| | ii) Hence find the set of values of k for which the equation $x^3 + 6x^2 + 9x + k = 0$ has three real and different roots. | 2 |
| c) | i) Find the domain and range of the function $f(x) = \tan^{-1}(e^x)$. | 1 |
| | ii) Sketch the curve $f(x) = \tan^{-1}(e^x)$ showing any intercepts on the coordinate axes and the equations of any asymptotes. | 2 |

Question 4. (15 marks) Start a new page.

Marks

- a) The ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where $a > b > 0$, has eccentricity $e = \frac{1}{2}$.
The point $P(2, 3)$ lies on the ellipse.

- i) Find the values of a and b . 3
- ii) Sketch the graph of the ellipse showing clearly the intercepts on the axes and the coordinates of the foci. 2

- b) The normal at the point $P\left(cp, \frac{c}{p}\right)$ on the hyperbola $xy = c^2$ meets the x -axis at Q . Also let M be the midpoint of PQ .

- i) Show that the normal at P has the equation $p^3x - py = c(p^4 - 1)$ 2
- ii) Show that M has coordinates $\left(\frac{c(2p^4 - 1)}{2p^3}, \frac{c}{2p}\right)$ 2
- iii) Hence or otherwise, find the equation of the locus of M . 3

- c) The polynomial $P(z)$ is defined by $P(z) = z^4 - 2z^3 - z^2 + 2z + 10$.

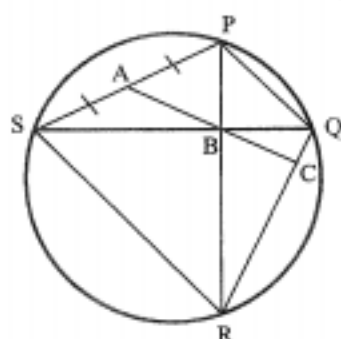
- i) Given that $z = 2 - i$ is a root of $P(z)$ write down another root giving a reason for your answer. 1
- ii) Hence, express $P(z)$ as a product of real quadratic factors. 2

Question 5. (15 marks) Start a new page.

Marks

- a) i) Suppose that the polynomial $P(x)$ has a double zero at $x = \alpha$.
Prove that $P'(x)$ also has a zero at $x = \alpha$. 2
- ii) The polynomial $P(x) = x^4 + ax^3 + bx + 21$ has a double zero at $x = 1$.
Find the values of a and b . 2
- b) i) The equation $x^3 + px^2 + qx + r = 0$ (where p, q, r are non zero) has
roots α, β, γ such that $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ are consecutive terms in an arithmetic
sequence. 3
- Show that $\beta = \frac{-3r}{q}$.
- ii) The equation $x^3 - 26x^2 + 216x - 576 = 0$ has roots α, β, γ such that
 $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ are consecutive terms in an arithmetic sequence. 3
- Find the values of α, β, γ .

c)



PQRS is a cyclic quadrilateral. The diagonals PR and SQ intersect at right angles at B. A is the midpoint of PS. AB produced meets QR at C.

Let $\angle ABP = \alpha$. Using the larger diagram provided to indicate angles, show that

- i) B, P and S are concyclic points. 1
- ii) $\angle APB = \angle ABP$. 1
- iii) AC is perpendicular to QR. 3

Question 6. (15 marks) Start a new page.

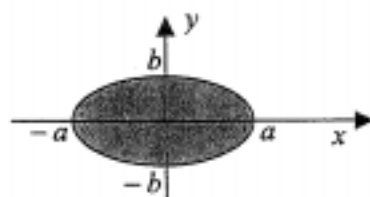
Marks

- a) If $\bar{z}_1 + \bar{z}_2 = 5 + 2i$, find $z_1 + z_2$ 1
- b) The arc of the curve $y = x(2 - x^2)$ from $x = 0$ to $x = 1$ is rotated about y axis. Find by using **cylindrical shells** the volume of the solid formed. 4
- c) i) Show that $a^2 + b^2 > 2ab$, where a and b are distinct positive real numbers. 1
- ii) Hence show that $a^2 + b^2 + c^2 > ab + bc + ca$, where a , b and c are distinct positive real numbers. 2
- iii) Hence or otherwise, prove that $\frac{a^2b^2 + b^2c^2 + c^2a^2}{a + b + c} > abc$. 2
- d) i) Prove that $\int_0^a f(x) dx = \int_0^a f(a - x) dx$ using the substitution $u = a - x$ 2
- ii) Hence evaluate $\int_0^2 x^2 \sqrt{2 - x} dx$, writing your answer in the form $a\sqrt{b}$. 3

Question 7. (15 marks) Start a new page.

Marks

a)

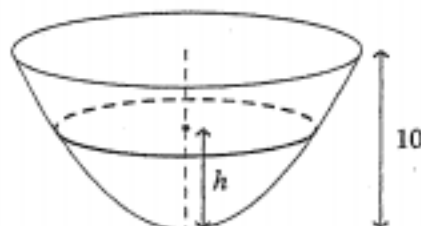


The diagram shows the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with major diameter $2a$ and minor diameter $2b$.

i) Show that the shaded area of the ellipse is given by $\frac{4b}{a} \int_0^a \sqrt{a^2 - x^2} dx$. 2

ii) Hence show that the shaded area is πab square units. 2

iii)



The diagram above shows a solid of height 10 cm. At height h cm above the vertex, the cross-section of the solid is an ellipse with major diameter $10\sqrt{h}$ cm and minor diameter $8\sqrt{h}$ cm.

α) Show that the cross-section at height h cm above the vertex has area $20\pi h$ cm². 2

β) Find the volume of the solid in exact form. 2

b) If α, β, γ are the roots of the equation $2x^3 - 7x^2 + 5x - 3 = 0$,

i) Show that the equation with roots $\alpha^2, \beta^2, \gamma^2$ is given by $4x^3 - 29x^2 - 17x - 9 = 0$ 2

ii) Hence evaluate $\alpha^3 + \beta^3 + \gamma^3$. 1

c) i) Expand $(\cos \theta + i \sin \theta)^3$ into powers of $\cos \theta$ and $\sin \theta$. 1

ii) By using De Moivre's Theorem show that $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$. 2

iii) Hence find the exact value of $4 \cos^3 \left(\frac{\pi}{12} \right) - 3 \cos \left(\frac{\pi}{12} \right)$. 1

Question 8. (15 marks) Start a new page.

Marks

- a) Find *all* solutions in radians of the equation

3

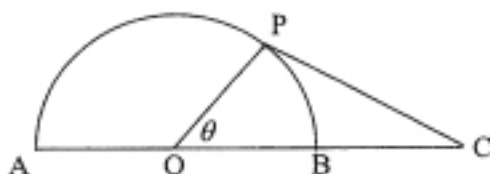
$$\frac{\sin^3 \theta + \cos^3 \theta}{\sin \theta + \cos \theta} = \frac{3}{4}$$

- b) For this question assume that tidal motion is simple harmonic.

On a certain day, the depth of water in a harbour at high tide at 5 am is 9 metres. At the following low tide at 11:20 am the depth is 3 metres. Find the latest time before noon that a ship can enter the harbour if a minimum depth of 7.5 metres is required. (Show all reasoning).

4

- c)



In the diagram above the fixed points A, O, B and C are on a straight line such that $AO = OB = BC = 1$ unit. The points A and B are also joined by a semicircle and P is a variable point on this semicircle such that $\angle POC = \theta$.

R is the region bounded by the arc AP of the semicircle and the straight lines AC and PC.

- i) Show that the area S of R is given by: $S = \frac{\pi}{2} - \frac{\theta}{2} + \sin \theta$. 1
- ii) Find the value of θ for which S is a maximum. 2
- iii) Show that the perimeter L of R is given by: 2

$$L = 3 + \pi - \theta + \sqrt{5 - 4 \cos \theta}.$$
- iv) Show that L has just one stationary point and that it occurs at the same value of θ for which S is a maximum. 2
- v) Hence find the greatest value of L in the interval $0 \leq \theta \leq \pi$. 1

END OF PAPER

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

CARINGBAH HIGH EXT 2 TRIAL SOLNS 2003

$$1a) I = \int \sec^2 x \tan^2 x dx + 2 \int \sec^2 x dx$$
$$= \int u^2 du + 2 \tan x \quad \{\text{where } u = \tan x\}$$

$$= \frac{1}{3} \tan^3 x + 2 \tan x + c$$

$$b) I = \int \frac{5}{(x+3)^2 + 4} dx$$
$$= 5/2 \tan^{-1}\left(\frac{x+3}{2}\right) + c$$

$$c) t = \tan \frac{x}{2} \Rightarrow dt = \frac{1}{2} \sec^2 \frac{x}{2} dx$$
$$\therefore dx = \frac{2dt}{1+t^2}$$

$$I = \int \frac{\frac{2}{1+t^2}}{\frac{1+\frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}}{1+t^2}} dt$$
$$= \int \frac{1}{1+t} dt$$
$$= \ln(1 + \tan \frac{x}{2}) + c$$

$$d) u = e^x + 1 \rightarrow du = e^x dx$$

$$I = \int \frac{e^x}{(e^x + 1)^2} \cdot e^x dx$$
$$= \int \frac{u-1}{u^2} du$$
$$= \int \frac{1}{u} - \frac{1}{u^2} du$$
$$= \ln(e^x + 1) + \frac{1}{e^x + 1} + c$$

$$e) I = \frac{1}{\ln 3} \cdot 3^x + c$$

9) i) Using parts with

$$u = x^n \quad v' = e^x$$
$$u' = n x^{n-1} \quad v = e^x$$

$$I_n = x^n e^x \Big|_0^1 - \int_0^1 (n x^{n-1}) e^x dx$$
$$= e - 0 - n \int x^{n-1} e^x dx$$
$$= e - n I_{n-1}$$

$$ii) I_3 = e - 3 I_2$$
$$= e - 3 [e - 2 I_1]$$
$$= e - 3e + 6 I_1$$
$$= -2e + 6 [e - I_0]$$

$$\text{where } I_0 = \int_0^1 e^x dx = e - 1$$

$$\therefore I_3 = -2e + 6 [e - (e - 1)]$$
$$= 6 - 2e$$

2a) i) $(3-4i)^2 = -7-24i$

ii) $i^3 \times \frac{3-4i}{2+5i} \times \frac{2-5i}{2-5i}$

$$= i^3 \left[\frac{-14-23i}{29} \right]$$

$$= \frac{-23}{29} + \frac{14i}{29}$$

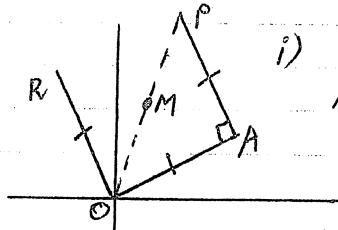
b) $a^2+b^2+i(a-b)=11+3i$
 $\therefore a^2+b^2+b=11$ } equating Real
 and $a=3$ } + Imag parts

$$\therefore b^2+b-2=0$$

$$(b+2)(b-1)=0$$

$$b=-2, b=1$$

$$\therefore z_1 = 3+i \quad z_2 = 3-2i$$

c)  i) $\vec{OR} = i\vec{OA} = iz$
 Also $\vec{OR} = \vec{AP}$
 $\vec{OP} = \vec{OA} + \vec{AP}$
 $= z + iz$
 $= (1+i)z$

ii) $\vec{OM} = \frac{1}{2} \vec{OP}$
 $= \frac{1+i}{2} z$

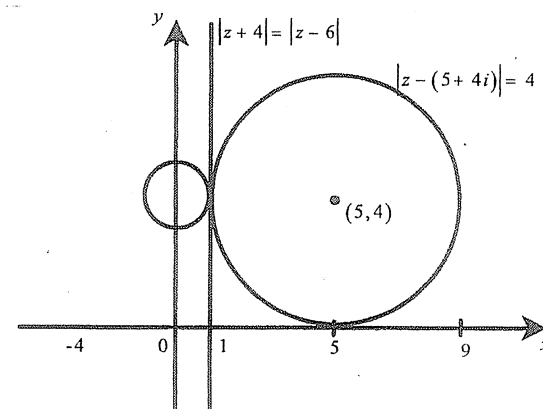
d) i) $|3-3i| = 3\sqrt{2} \quad \arg(3-3i) = -\frac{\pi}{4}$

$$\therefore 3-3i = 3\sqrt{2} (\cos(-\pi/4) + i\sin(-\pi/4))$$

$$= 3\sqrt{2} (\cos(\pi/4) - i\sin(\pi/4))$$

ii) $\therefore (3-3i)^7 = (3\sqrt{2})^7 (\cos \frac{7\pi}{4} - i\sin \frac{7\pi}{4})$
 $= 17496\sqrt{2} [\frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}]$
 $= 17496 + 17496i$

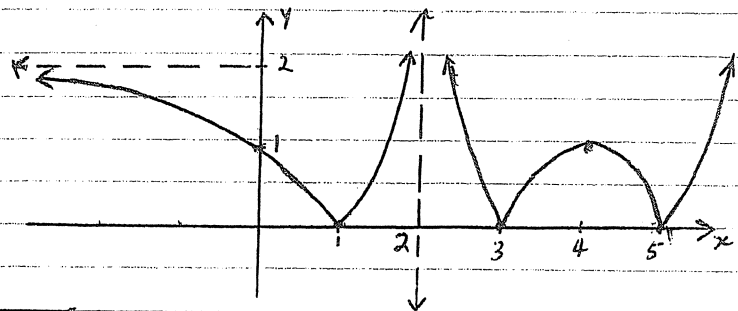
e) i)



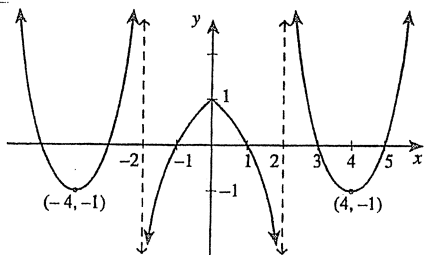
ii) $z = 1+4i$

iii) $k=1$

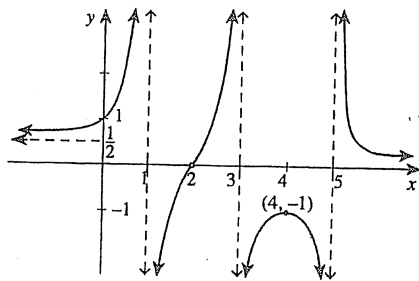
3a) i)



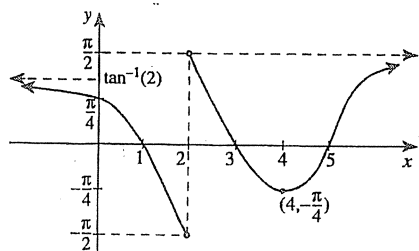
ii)



iii)



iv)



3b) i) $y = x^3 + 6x^2 + 9x + k$

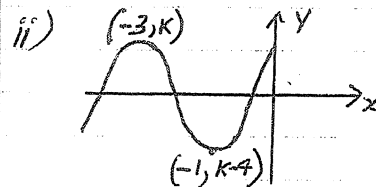
$$y' = 3x^2 + 12x + 9$$

$$y'' = 6x + 12$$

$$y' = 0 \rightarrow x = -1, -3$$

$x = -1, y = k - 4, y'' = 6 > 0 \therefore (-1, k - 4)$ is a minimum.

$x = -3, y = k, y'' = -6 < 0 \therefore (-3, k)$ is a maximum.

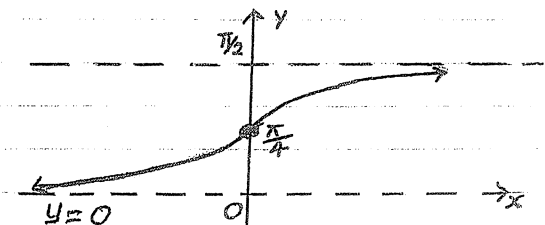


To have 3 real and different roots
the curve must cut the x -axis in
3 distinct points. Hence $k, k - 4$
must have opposite signs
 $\therefore 0 < k < 4$

c) $f(x) = \tan^{-1}(e^x)$

D: all real x

$$R: 0 < y < \pi/2$$



4a) i) $e = 1/2 \therefore b^2 = a^2(1 - \frac{1}{4})$

$\therefore b^2 = \frac{3}{4}a^2$

Since $P(2,3)$ lies on E , then

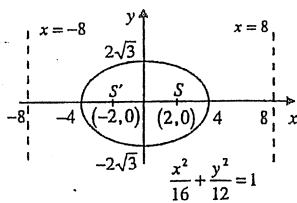
$$\frac{4}{a^2} + \frac{9}{b^2} = 1$$

$$\therefore \frac{4}{a^2} + \frac{12}{a^2} = 1$$

$$\therefore a^2 = 16 \quad \& \quad b^2 = 12$$

$$\therefore a = 4 \quad b = 2\sqrt{3}$$

ii)



b) i) $y = c^2/x \Rightarrow \frac{dy}{dx} = -\frac{c^2}{x^2}$

$\therefore$ At $x = cp$ grad. of Tangent $= -1/p^2$

$\therefore$ grad. of Normal $= p^2$

$\therefore$ eqⁿ of N: $y - \frac{c}{p} = p^2(x - cp)$

$$py - c = p^3x - cp^4$$

$$\therefore p^3x - py = c(p^4 - 1)$$

ii) At Q $y=0 \Rightarrow p^3x = c(p^4 - 1)$
 $x = \frac{c(p^4 - 1)}{p^3}$

$$\therefore M = \left\{ \frac{\frac{c(p^4 - 1)}{p^3} + cp}{2}, \frac{\frac{c}{p} + 0}{2} \right\}$$

$$= \left\{ \frac{c(2p^4 - 1)}{2p^3}, \frac{c}{2p} \right\}$$

iii)

$y = c/2p \Rightarrow p = c/2y \rightarrow p^3 = \frac{c^3}{8y^3} \quad \& \quad p^4 = \frac{c^4}{16y^4}$

$$\therefore x = \frac{c}{2} \left[\frac{2 \times \frac{c^4}{16y^4} - 1}{\frac{c^3}{8y^3}} \right]$$

$$= \frac{c}{2} \left[\frac{2c^4 - 16y^4}{16y^4} \right] \times \frac{8y^3}{c^3}$$

$$= \frac{c^4 - 8y^4}{2c^2y}$$

$\therefore$ Locus is $2c^2yx = c^4 - 8y^4$

c) i) $z = 2+i$ { roots occur in complex conjugate pairs when coeff's are real.

ii) $P(z) = [z - (2+i)][z - (2-i)]Q(z)$

$$= (z^2 - 4z + 5)Q(z)$$

$$= (z^2 - 4z + 5)(z^2 + 2z + 2)$$

5a)

i) $P(x) = (x-\alpha)^2 Q(x)$ (where $Q(x)$ is a polynomial)

$$P'(x) = 2(x-\alpha)Q(x) + (x-\alpha)^2 Q'(x) \\ = (x-\alpha)[2Q(x) + (x-\alpha)Q'(x)]$$

$$\therefore P'(\alpha) = 0 \quad \text{i.e. } x = \alpha \text{ is a root of } P'(x) = 0$$

ii) $P(x) = x^4 + ax^3 + bx + 21$

$$P'(x) = 4x^3 + 3ax^2 + b$$

$$P(1) = 1 + a + b + 21 = 0$$

$$P'(1) = 4 + 3a + b = 0$$

$$a + b = -22$$

$$3a + b = -4$$

Solving simultaneously $a = 9, b = -31$.

b)

(i) $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ in AP $\Rightarrow \frac{1}{\beta} - \frac{1}{\alpha} = \frac{1}{\gamma} - \frac{1}{\beta} \Rightarrow \frac{2}{\beta} = \frac{1}{\alpha} + \frac{1}{\gamma}$. Then $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{3}{\beta}$
But $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \gamma\alpha + \alpha\beta}{\alpha\beta\gamma} = -\frac{q}{r}$. Hence $\frac{3}{\beta} = -\frac{q}{r} \therefore \beta = \frac{-3r}{q}$

(ii) $x^3 - 26x^2 + 216x - 576 = 0$ such that $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ in AP $\Rightarrow \beta = \frac{-3r}{q} = \frac{3 \times 576}{216} = 8$

$$\text{Then } \alpha + \gamma = 26 - 8 = 18 \text{ and } \alpha\gamma = 576 \div 8 = 72.$$

$$\text{Hence } \alpha, \gamma \text{ zeros of } x^2 - 18x + 72 = (x-12)(x-6).$$

α, β, γ are 6, 8, 12 or 12, 8, 6 respectively.

c)

i) $\angle PBS = 90^\circ$ {diagonals intersect at right $\angle$ }.
 $\therefore \angle$ in a semi-circle with PS the diameter.

ii)

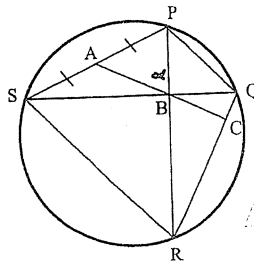
Since A is the midpt of diameter PS
then $AP = AB$ {= radii} $\rightarrow \Delta APB$ isos
Hence $\angle APB = \angle ABP$

iii) $\angle ABS = 90 - \alpha \rightarrow \angle QBC = 90 - \alpha$ {vertically opp}

$\angle SPR = \angle SQR = \alpha$ { $\angle$ in same segment}

$$\therefore \angle BCQ = 180 - (\alpha) - (90 - \alpha) \\ = 90^\circ$$

$\therefore AC \perp QR$



6) a) $\bar{z}_1 + \bar{z}_2 = \overline{z_1 + z_2} \Rightarrow z_1 + z_2 = 5 - 2i$

b)

$$\Delta V = 2\pi xy \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^1 2\pi xy \Delta x$$

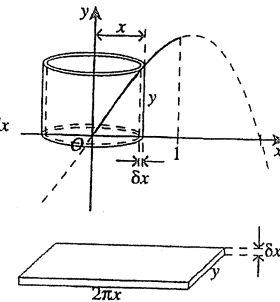
$$= \int_0^1 (2\pi x) x (2 - x^2) dx$$

$$= 2\pi \left[\frac{2x^3}{3} - \frac{x^5}{5} \right]_0^1$$

$$= 2\pi \left[\frac{2}{3} - \frac{1}{5} \right]$$

$$= 2\pi \left[\frac{7}{15} \right]$$

$$= \frac{14\pi}{15} \text{ cubic units.}$$



c) i)

Since $a \neq b$, $a - b > 0$, so $(a - b)^2 > 0$.

$$\text{Hence } a^2 - 2ab + b^2 > 0$$

$$a^2 + b^2 > 2ab.$$

ii)

$$\text{From (i), } a^2 + b^2 > 2ab,$$

$$b^2 + c^2 > 2bc \text{ and}$$

$$a^2 + c^2 > 2ac.$$

$$\text{Adding, } 2(a^2 + b^2 + c^2) > 2(ab + ac + bc)$$

$$a^2 + b^2 + c^2 > ab + ac + bc.$$

iii)

Let $A = ab$, $B = bc$ and $C = ac$.

Then A, B and C are distinct positive numbers, and from (ii),

$$A^2 + B^2 + C^2 > AB + AC + BC.$$

Substituting,

$$a^2b^2 + b^2c^2 + a^2c^2 > (ab)(bc) + (ab)(ac) + (bc)(ac).$$

$$\text{Now } (ab)(bc) + (ab)(ac) + (bc)(ac) = abc(a + b + c).$$

$$\text{Hence } \frac{a^2b^2 + b^2c^2 + a^2c^2}{a + b + c} > abc.$$

6d) i) $u = a - x$
 $\therefore du = -dx$

When $x = 0 \rightarrow u = a$
 $x = a \rightarrow u = 0$

$$\begin{aligned} \therefore \text{LHS} &= \int_0^a f(x) dx \\ &= -\int_a^0 f(a-u) du \\ &= \int_0^a f(a-u) du \\ &= \text{RHS} \end{aligned}$$

$$\begin{aligned} \text{ii)} \int_0^2 x^2 \sqrt{2-x} dx &= \int_0^2 (2-x)^2 \cdot \sqrt{x} dx \\ &= \int_0^2 4x^{1/2} - 4x^{3/2} + x^{5/2} dx \\ &= \left[\frac{8}{3} x^{3/2} - \frac{8}{5} x^{5/2} + \frac{2}{7} x^{7/2} \right]_0^2 \\ &= \frac{8}{3} (\sqrt{2})^3 - \frac{8}{5} (\sqrt{2})^5 + \frac{2}{7} (\sqrt{2})^7 - 0 \\ &= \frac{128\sqrt{2}}{105} \end{aligned}$$

Q7a) i) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\begin{aligned} y &= \pm b \sqrt{1 - \frac{x^2}{a^2}} \\ &= \pm \frac{b}{a} \sqrt{a^2 - x^2} \end{aligned}$$

Area of the ellipse in the first quadrant $= \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} dx$

$\therefore$ Area of the ellipse $= \frac{4b}{a} \int_0^a \sqrt{a^2 - x^2} dx$

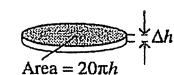
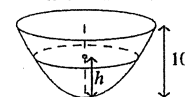
ii) $y = \sqrt{a^2 - x^2}$ is the equation of a circle with centre at the origin and radius a .

The expression $\int_0^a \sqrt{a^2 - x^2} dx$ gives the area of the first

quadrant of this circle, which is equal to $\frac{\pi a^2}{4}$.

Hence the area of the ellipse is $\frac{4b}{a} \frac{\pi a^2}{4} = \pi ab$.

iii) α . $a = 4\sqrt{h}$ and $b = 5\sqrt{h}$.
 From (ii),
 area $= \pi ab$
 $= \pi (4\sqrt{h})(5\sqrt{h})$
 $= 20\pi h$



β . Volume $= \int_0^{10} 20\pi h dh$
 $= [10\pi h^2]_0^{10}$
 $= 1000\pi \text{ cm}^3$

7b) i) Let $x = \alpha^2 \rightarrow \alpha = \sqrt{x}$

$\therefore 2(\sqrt{x})^3 - 7(\sqrt{x})^2 + 5\sqrt{x} - 3 = 0$

$2x\sqrt{x} - 7x + 5\sqrt{x} - 3 = 0$

$\sqrt{x}(2x+5) = 7x+3$

$x(4x^2+20x+25) = 49x^2+42x+9$

$\therefore 4x^3 - 29x^2 - 17x - 9 = 0$

ii) α, β, γ are the roots of:

$2x^3 - 7x^2 + 5x - 3 = 0$

$\therefore 2\alpha^3 = 7\alpha^2 - 5\alpha + 3$

$2\beta^3 = 7\beta^2 - 5\beta + 3$

$2\gamma^3 = 7\gamma^2 - 5\gamma + 3$

$\therefore 2(\alpha^3 + \beta^3 + \gamma^3) = 7(\alpha^2 + \beta^2 + \gamma^2) - 5(\alpha + \beta + \gamma) + 9$

$\therefore \alpha^3 + \beta^3 + \gamma^3 = \frac{1}{2} \left[7 \times \frac{29}{4} - 5 \times \frac{7}{2} + 9 \right]$

$= 21\frac{1}{8} \text{ or } \frac{169}{8}$

c) i) $(\cos\theta + i\sin\theta)^3 = \cos^3\theta + 3\cos^2\theta(i\sin\theta) + 3\cos\theta(i\sin\theta)^2 + (i\sin\theta)^3$
 $= \cos^3\theta - 3\cos\theta\sin^2\theta + i(3\cos^2\theta\sin\theta - \sin^3\theta) \quad \text{--- (1)}$

ii) $(\cos\theta + i\sin\theta)^3 = \cos 3\theta + i\sin 3\theta \quad \text{--- (2)}$

$\therefore$ equating real parts of (1) & (2)

$\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta$

$= \cos^3\theta - 3\cos\theta(1 - \cos^2\theta)$

$= 4\cos^3\theta - 3\cos\theta$

iii) $\cos\left(3 \times \frac{\pi}{12}\right) = \frac{1}{\sqrt{2}}$

8a) $\frac{s^3 + c^3}{s+c} = \frac{(s+c)(s^2 - sc + c^2)}{s+c}$
 $= 1 - sc$

$\therefore 1 - \sin\theta\cos\theta = \frac{3}{4}$

$\sin\theta\cos\theta = \frac{1}{4}$

$\sin 2\theta = \frac{1}{2}$

$2\theta = n\pi + (-1)^n \left(\frac{\pi}{6}\right)$

$\therefore \theta = \frac{n\pi}{2} + (-1)^n \frac{\pi}{12}$

b) High Tide = 9m at 5am. // Low T = 3m at 11:20am
 $\therefore$ amplitude = $\frac{9-3}{2} = 3\text{m}$

$P = \frac{2\pi}{n} \therefore 760 = 2\pi/n \rightarrow n = \frac{\pi}{380}$

$t=0 \quad \text{---} \quad 3$

$\text{---} \quad 1.5$

$\text{---} \quad 0$

$\text{---} \quad -3$

Since the motion is SHM & periodic then $\ddot{x} = -n^2x$ which has solution

$x = a\cos(nt + \alpha)$

$\therefore x = 3\cos\left(\frac{\pi t}{380} + \alpha\right)$

When $x=3, t=0 \Rightarrow \alpha=0$

$\therefore x = 3\cos\left(\frac{\pi t}{180}\right)$

When $x=1.5 \Rightarrow 0.5 = \cos\frac{\pi t}{380}$

$\therefore t = \frac{380}{\pi} \times \cos^{-1}(0.5) \doteq 126\text{min} (2\text{h } 6\text{min})$

$\therefore$ the latest time before noon is 7:06am.

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i) Area of $\triangle OPC = \frac{1}{2} \times 1 \times 2 \times \sin \theta = \sin \theta$

Area of sector OPB = $\frac{1}{2} \times 1 \times 1 \times \theta = \theta/2$

Area of semi-circle = $\frac{1}{2} \times \pi \times 1^2 = \pi/2$

$\therefore$ Area of S = Area of $\triangle OPC$ + semi-circle - sector OPB
 $= \pi/2 - \theta/2 + \sin \theta$

ii) $S' = \cos \theta - 1/2$ & $S'' = -\sin \theta$

$S' = 0 \rightarrow \theta = \pi/3$ & $S''(\pi/3) = -\sqrt{3}/2 < 0$

$\therefore$ Max S at $\theta = \pi/3$.

iii) $L = AP + AC + PC = (AB - PB) + AC + PC$

Now $PC^2 = 1^2 + 2^2 - 2 \times 1 \times 2 \times \cos \theta = 5 - 4 \cos \theta$

$\therefore PC = \sqrt{5 - 4 \cos \theta}$

$AP = \frac{2\pi \times 1}{2} - r\theta = \pi - \theta$

$AC = 3$

$\therefore L = 3 + \pi - \theta + \sqrt{5 - 4 \cos \theta}$

iv) $L' = -1 + \frac{1}{2} (5 - 4 \cos \theta)^{-1/2} \times 4 \sin \theta$
 $= -1 + \frac{2 \sin \theta}{\sqrt{5 - 4 \cos \theta}}$

$\therefore L' = 0 \Rightarrow \sqrt{5 - 4 \cos \theta} = 2 \sin \theta$

$5 - 4 \cos \theta = 4 \sin^2 \theta$

$5 - 4 \cos \theta = 4(1 - \cos^2 \theta)$

$\therefore (2 \cos \theta - 1)^2 = 0 \Rightarrow \cos \theta = 1/2 \Rightarrow \theta = \pi/3$

Hence only 1 stat pt at $\theta = \pi/3$.

v) TESTING L' either side of $\theta = \pi/3$

θ	$\pi/6$	$\pi/3$	$\pi/2$
L'	-	0	-

gives a Horizontal Point of inflexion + shows the

curve is a decreasing function

$\therefore$ greatest value occurs when $\theta = 0$

$\therefore L_{\max} = 4 + \pi$